

Infosys
Springboard

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AirFly Insights: Data Visualization and Analysis of Airline Operations Report

Presented by: Anitha Chava

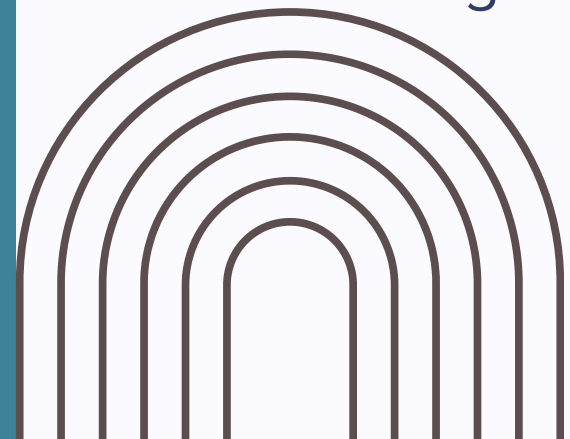
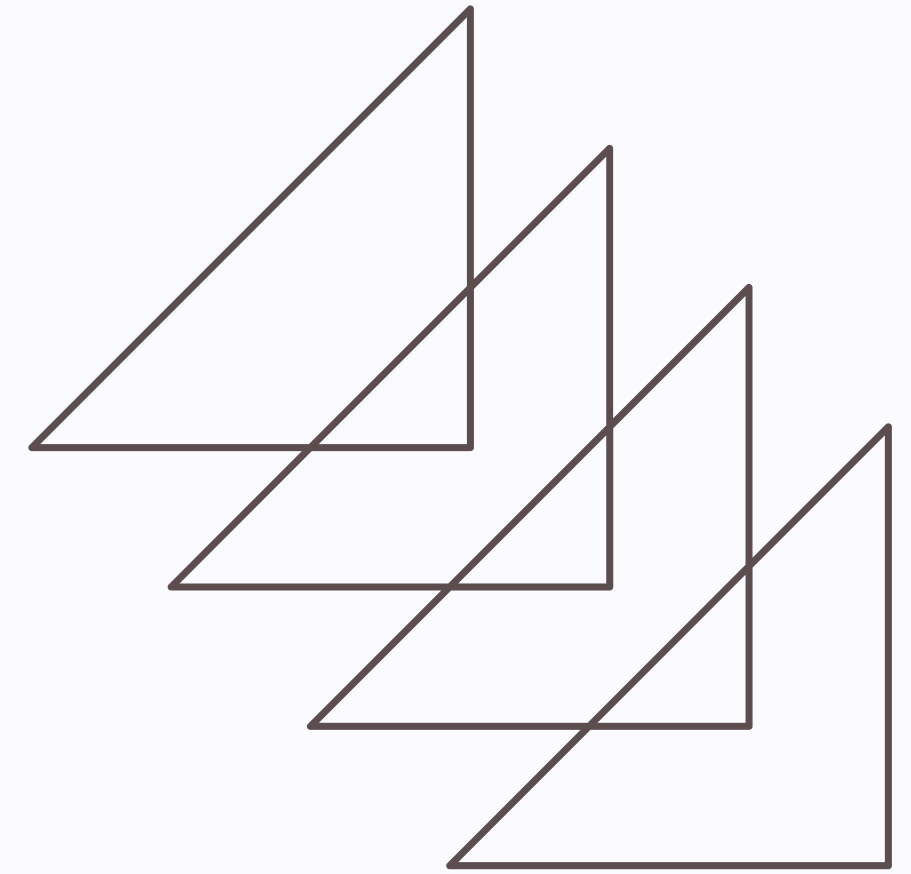
Mentored by: Rahul Yadav



Introduction

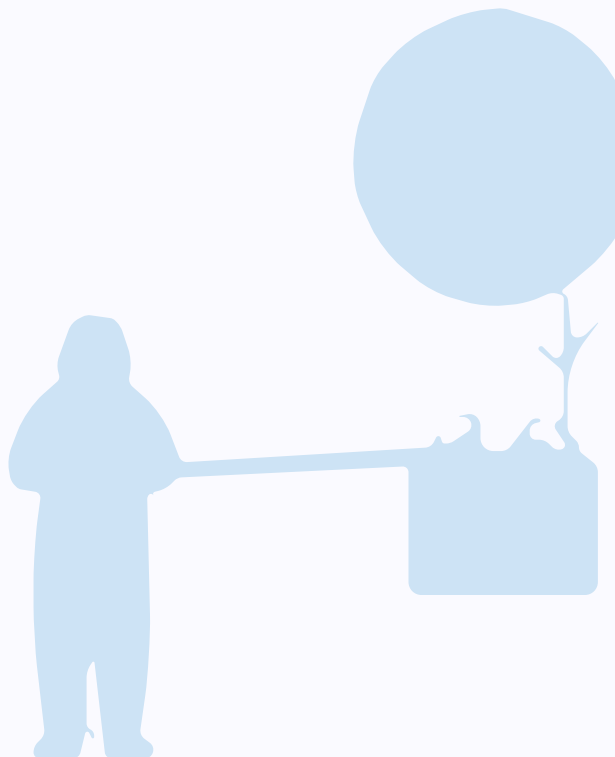
Overview of objectives and problem statement:

- Analyze large-scale airline flight data to identify operational trends
- Study flight delay patterns across airlines and airports
- Examine common reasons for flight cancellations
- Use data visualization techniques to represent insights clearly
- Evaluate airline-level and airport-level performance
- Support data-driven and actionable insights through visual analysis





Expected Outcomes

- Understand and preprocess aviation datasets for analysis
 - Explore trends in flight schedules, delays, cancellations, and routes
 - Visualize key metrics using bar charts, time series, heatmaps, maps, and comparisons
 - Provide insights for stakeholders including airline operators and analysts
 - Summarize findings through a final visual report and presentation
- 

Project Workflow & Objectives



Understand & Preprocess Aviation Datasets for Analysis



Explore Trends in Flight Schedules, Delays, Cancellations, & Routes



Visualize Key Metrics Using Bar Charts, Time Series, Heatmaps, Maps, & Comparisons



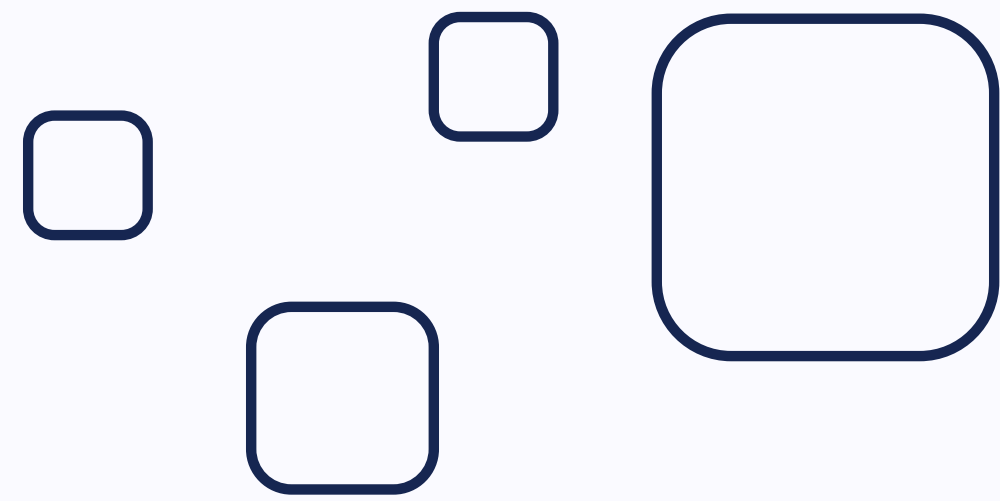
Provide Insights for Stakeholders Including Airline Operators & Analysts



Summarize Findings through Final Visual Report & Presentation

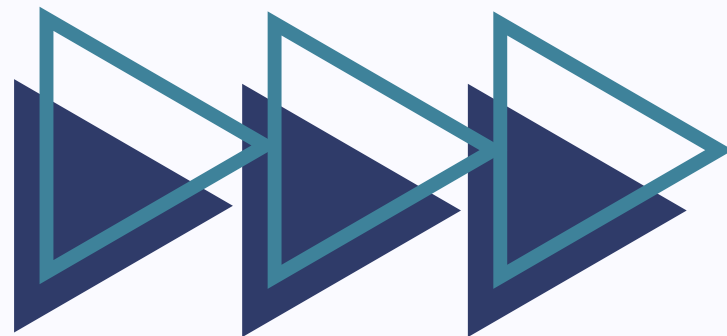
Expected Outcomes:

- Actionable Insights for Operational Improvement
 - Comprehensive Visual Report for Stakeholders
- 



Dataset Overview

Insight into the data source, size,
and key variables analyzed



Source: Kaggle

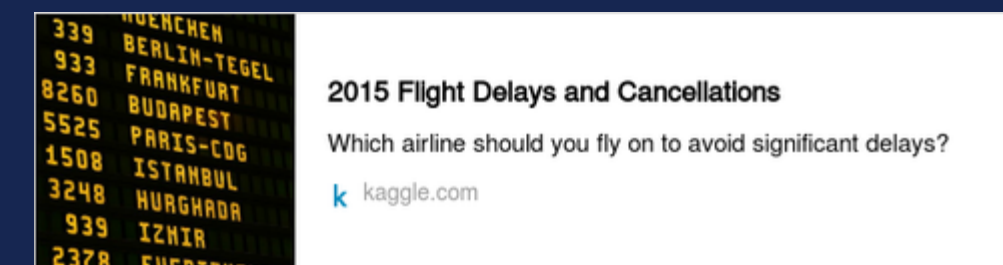


Datasets and Size

1. Flights (5819079 rows, 31 cols)
2. Airports (322 rows, 7 cols)
3. Airlines (14 rows, 2 cols)



Link:

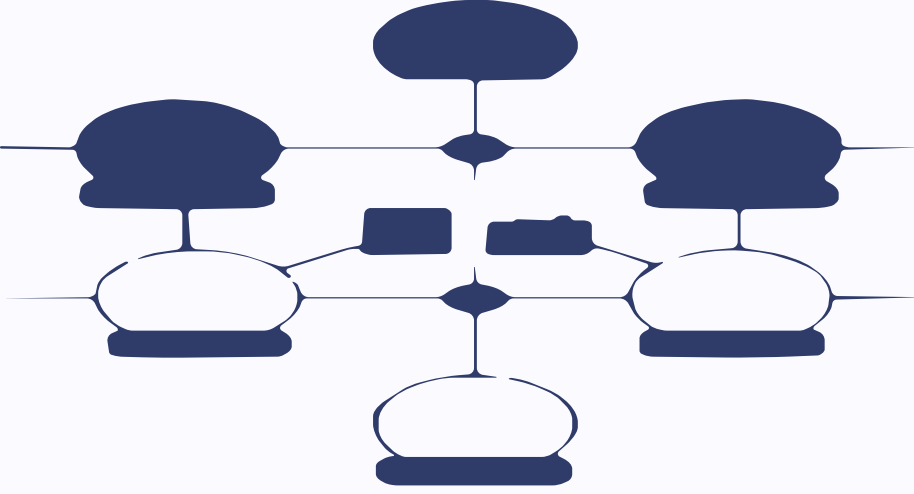


Methodology Overview

Key steps taken in cleaning data and analytical approach utilized

- Data Acquisition and Understanding: Collected and reviewed flight datasets.
- Data Cleaning and Feature Engineering: Cleaned data and created key features.
- Univariate and Bivariate Analysis: Analyzed metrics and relationships.
- Delay Cause Analysis: Identified main delay reasons.
- Cancellation and Seasonal Trends: Studied seasonal cancellation patterns.
- Airport and Route-Level Exploration: Compared airports and routes.
- Final Dashboard or Report: Visualized insights clearly.
- Documentation and Presentation: Documented and presented results.

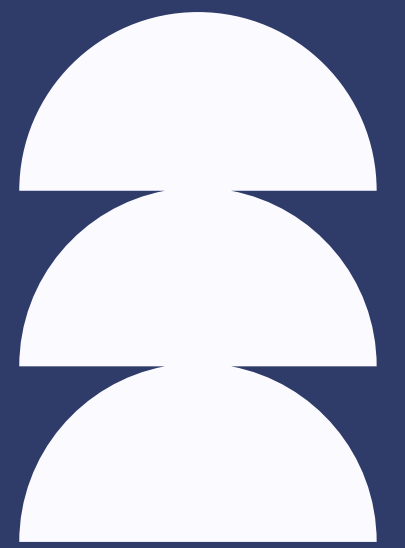




Project Workflow



Exploratory Analysis

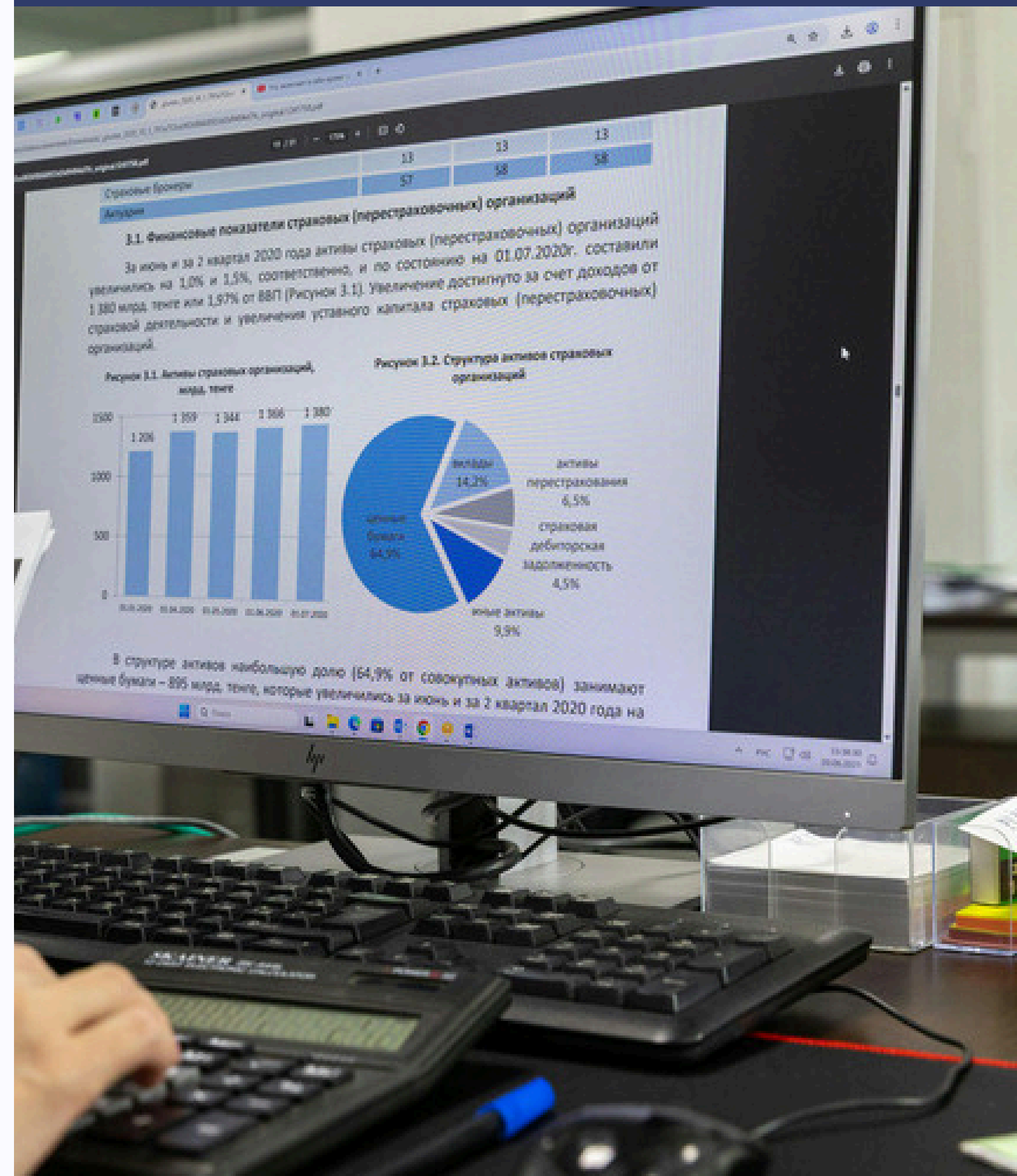


Key Patterns and Insights from Data Exploration Techniques

- Defined project goals, and overall workflow
- Loaded CSV datasets using pandas
- Explored schema, data types, size, and null values
- Performed data sampling and memory optimization
- Handled null values in delay and cancellation columns
- Created derived features such as month, day, hour, and route
- Formatted and standardized datetime columns
- Performed Feature engineering, Ex: Date, Route etc

Visualization Highlights

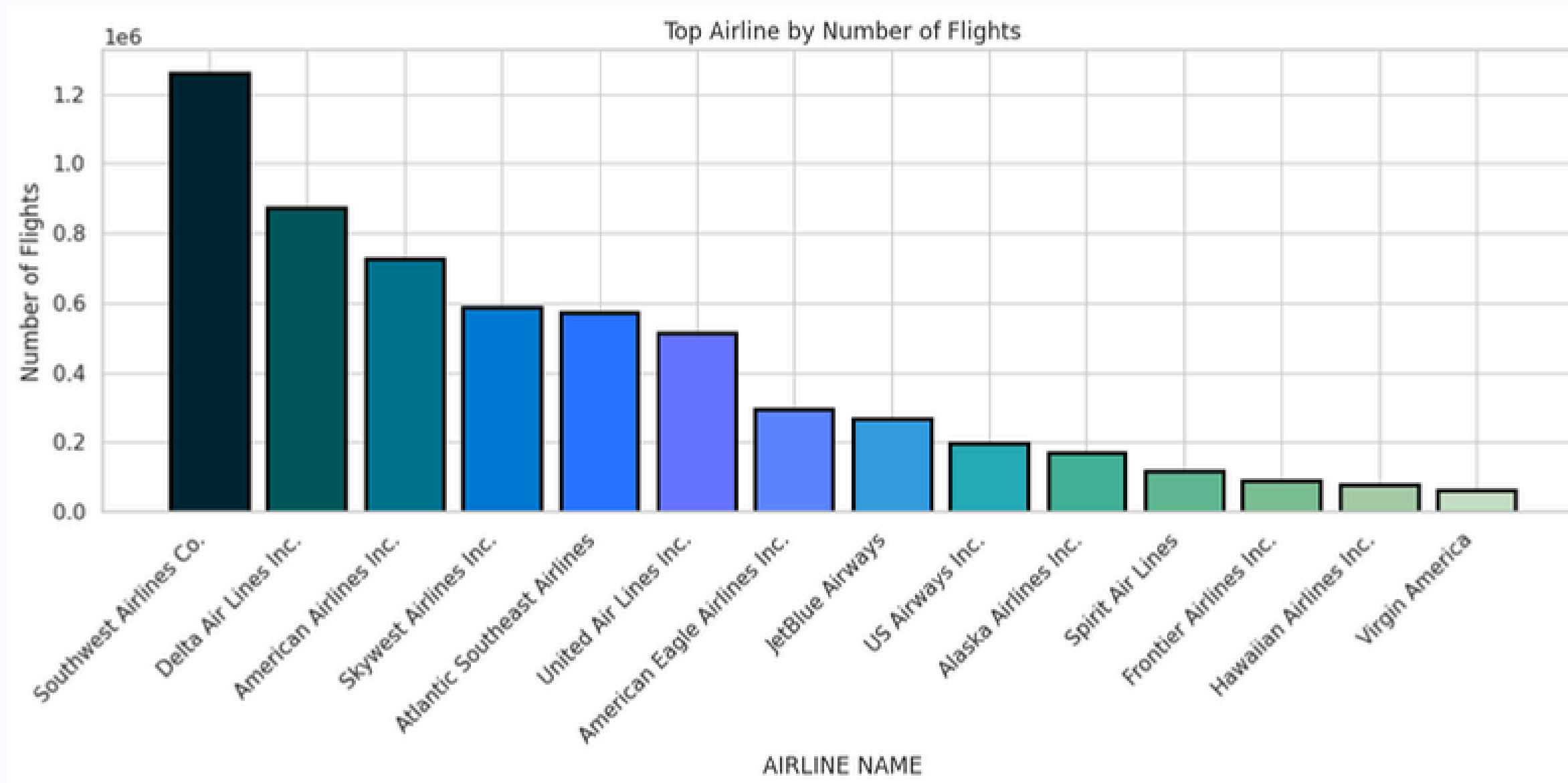
Next Milestones focused on visual analysis to uncover delay, route, and cancellation patterns across airlines, airports, and time periods. The analysis highlighted peak delay times, delay-prone carriers, congested routes, and strong seasonal effects on cancellations, supported through multiple data visualizations.



Uni & Bivariate Visuals

& thier Findings:

Top Airlines

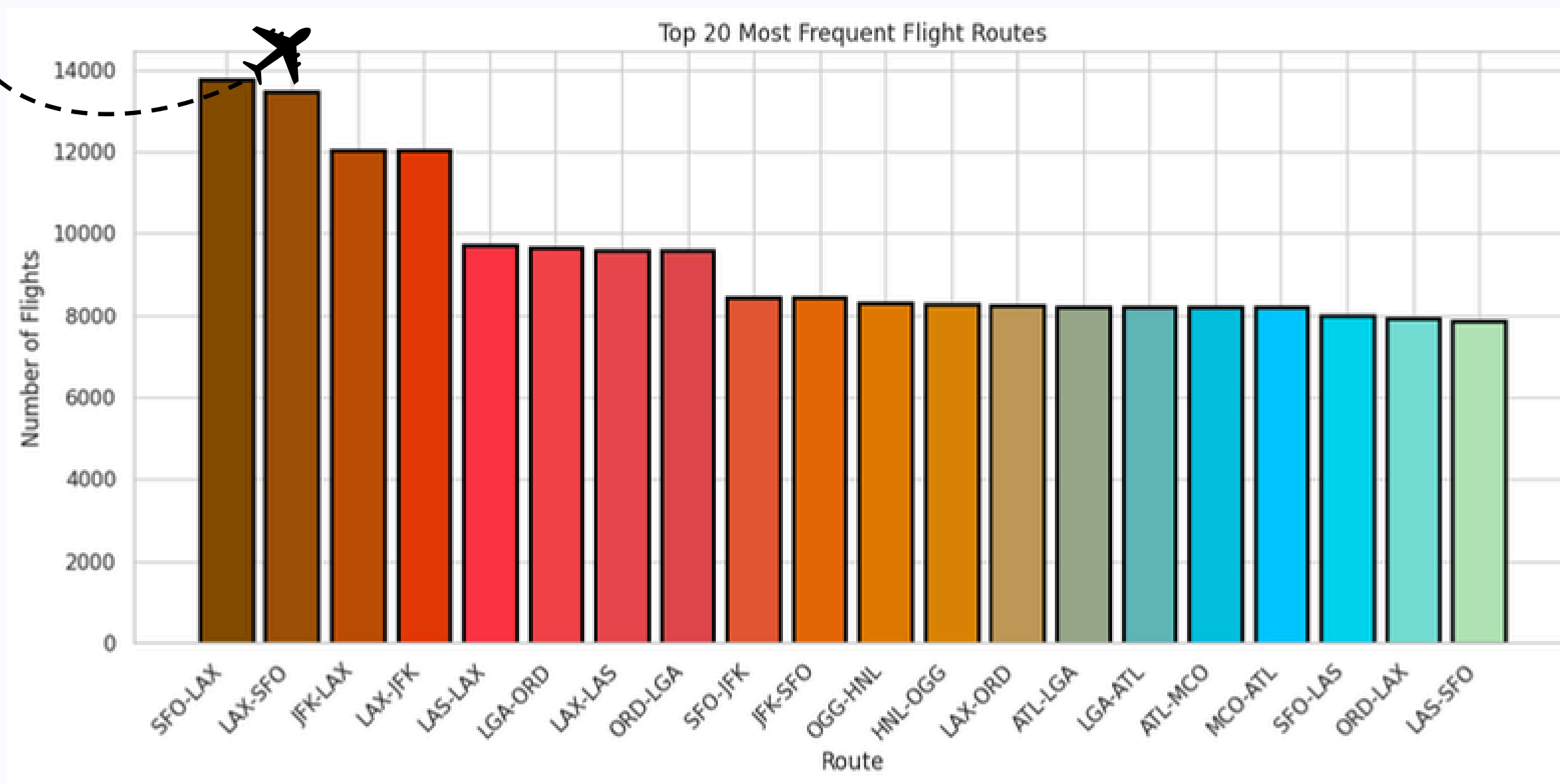


Business Context

Southwest (WN) is the busiest airline, followed by Delta and American, with the top five carriers handling over 65% of all flights. Smaller airlines contribute minimally, so operational issues with major carriers have the greatest system-wide impact.



Top Routes



Insights and Suggestions:

- High-traffic routes like LAX–SFO and JFK–LAX dominate operations, indicating strong demand on major hub-to-hub corridors and key revenue-generating routes.
- Optimizing scheduling, capacity, and delay management on these routes can significantly improve efficiency, punctuality, and overall profitability.



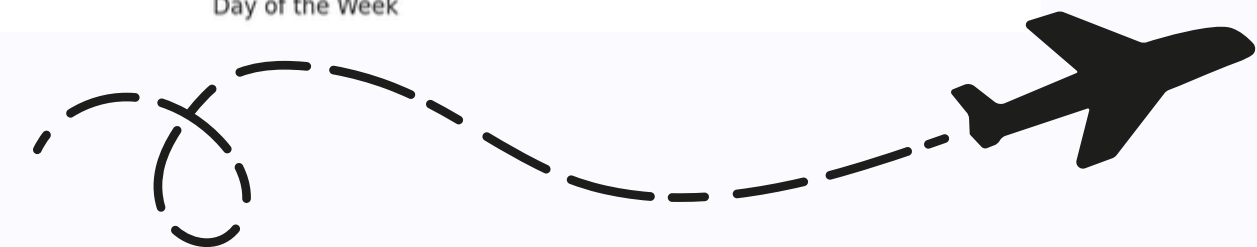
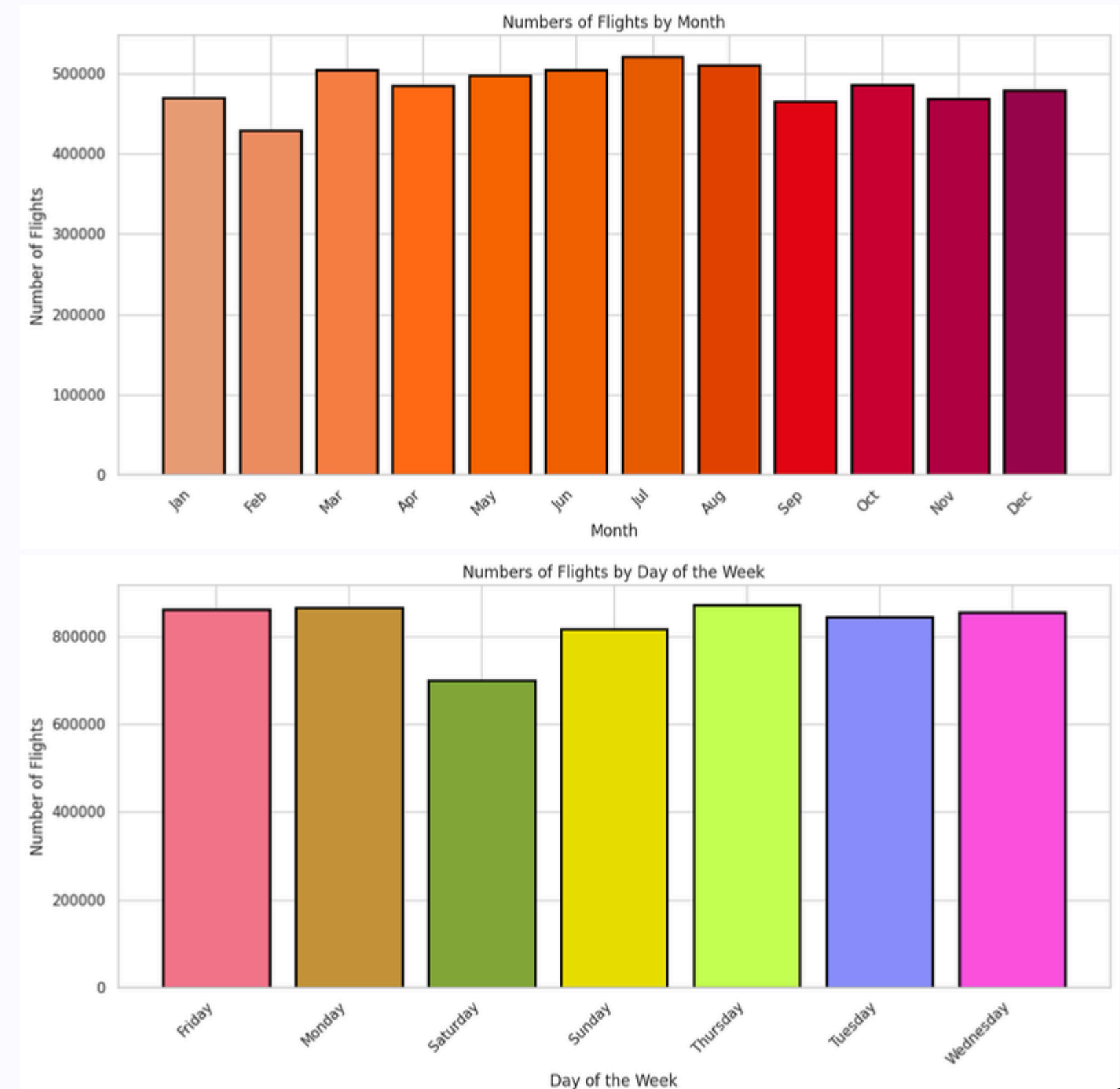
Flight Distribution by Month & Day

Flights by Month:

- Flights peak in mid-year, especially July and August.
- Early and late months show relatively lower traffic.
- Plan resources and pricing for higher summer demand.

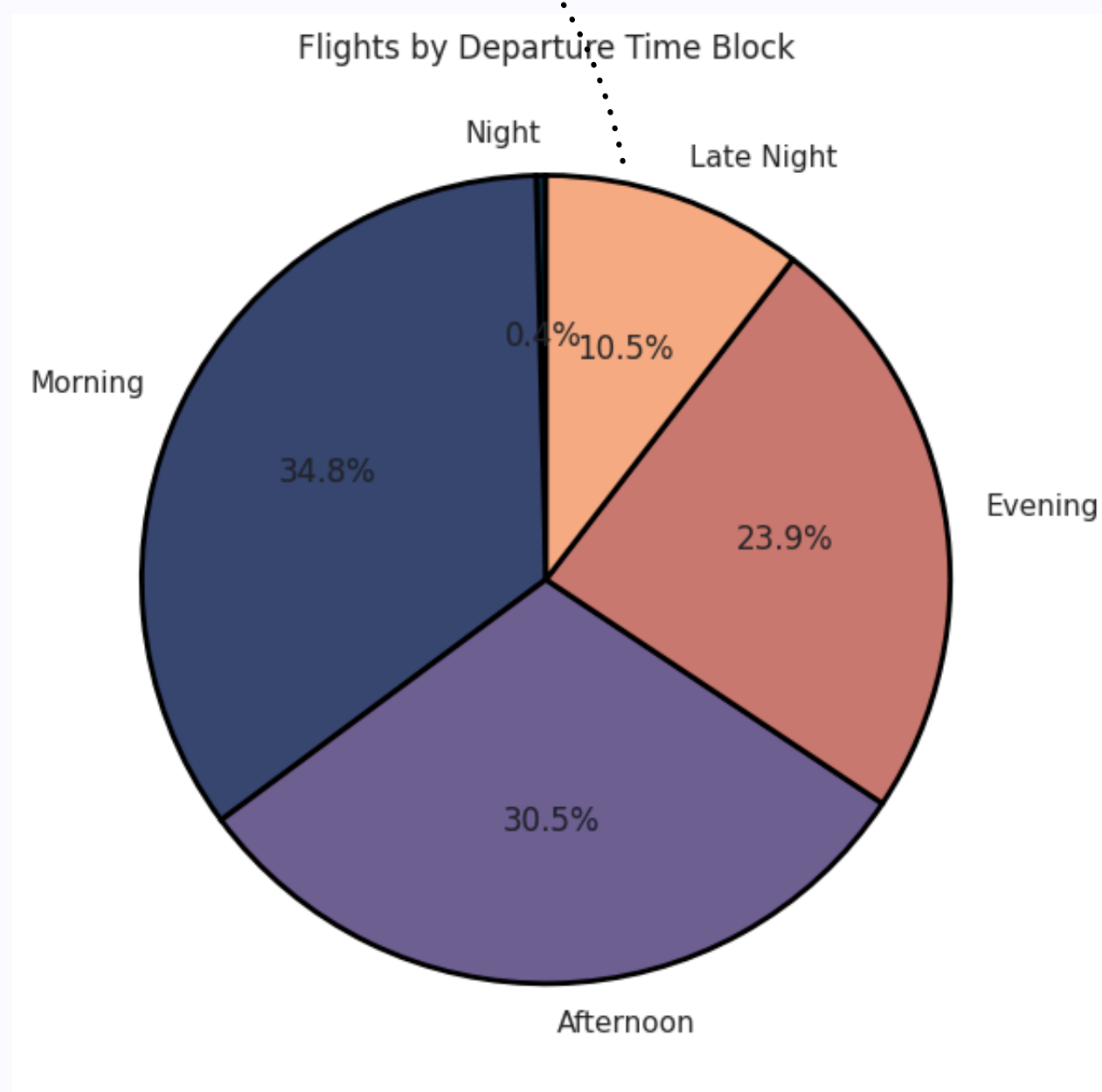
Flights by Day:

- Weekdays have higher flight volumes.
- Saturday is the least busy day.
- Optimize schedules and maintenance on weekends.





Flights by Departure Time



Findings

- Most flights operate in the morning and afternoon, together accounting for nearly two thirds of total traffic.
- Evening flights have moderate demand, while late night and night flights are very low.

Suggestions to Improve Business

- Increase capacity, pricing optimization, and premium services during morning and afternoon peak hours.
- Use discounts or special offers to boost demand for evening and late-night flights.
- Schedule maintenance or low priority routes during night hours to reduce operating costs.

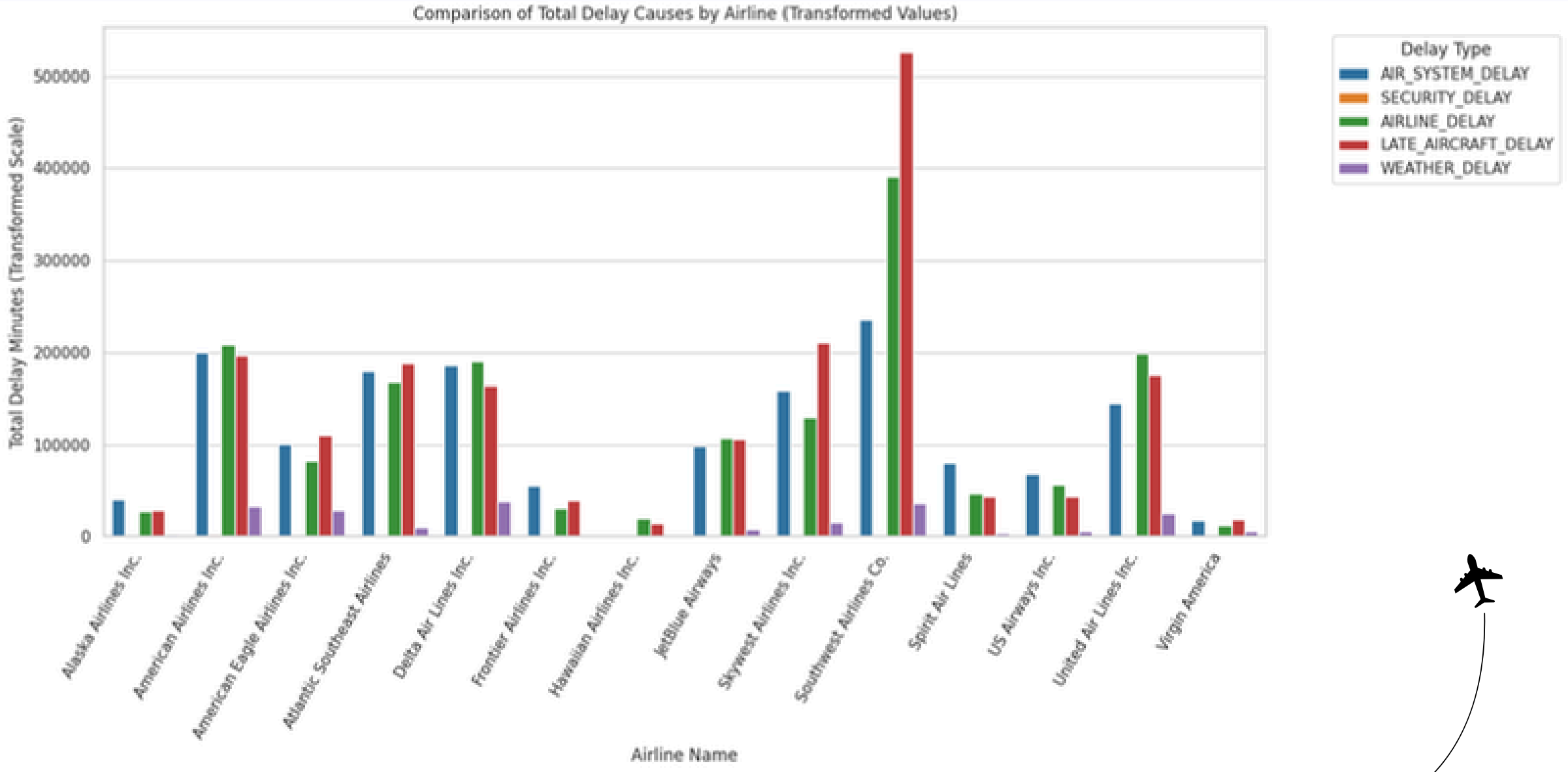
Comparision of Delay Causes by Airline

Findings

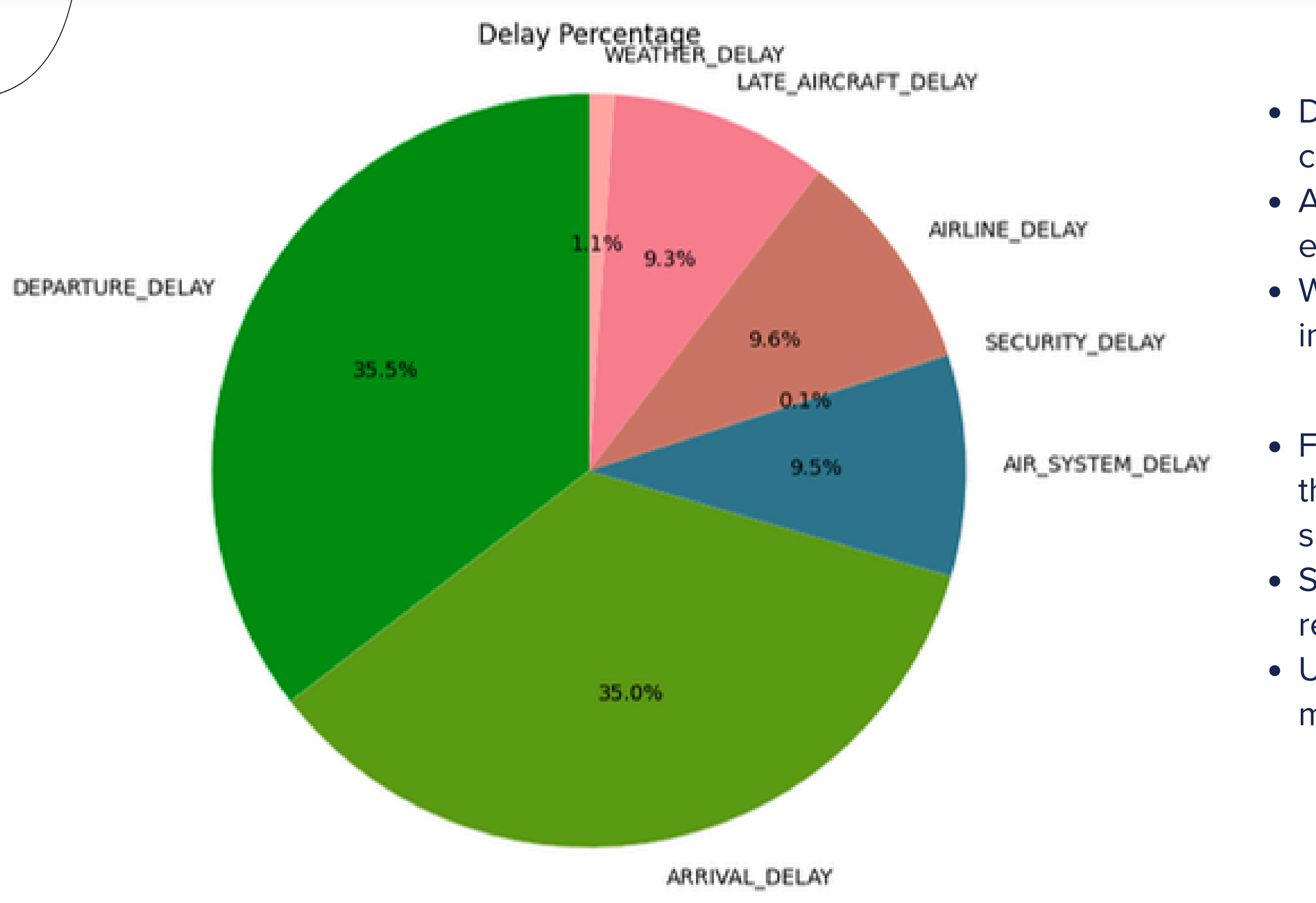
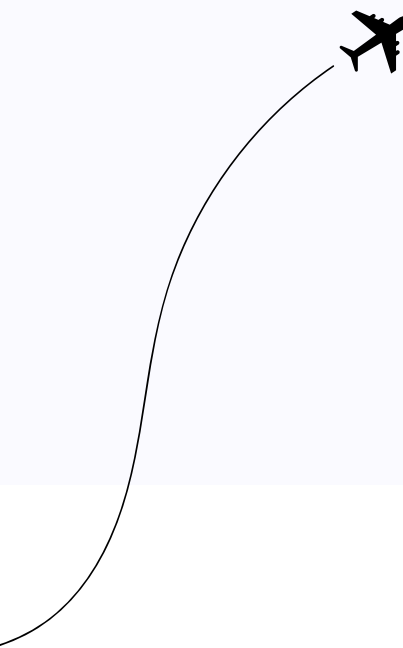
- Late aircraft and airline-related delays are the largest contributors across most airlines.
- Major carriers show significantly higher total delays due to higher flight volumes.
- Weather and security delays are comparatively low and less impactful.

Suggestions to Improve

- Improve aircraft turnaround time and crew scheduling to reduce late aircraft delays.
- Use predictive maintenance and real-time operations tracking to minimize airline delays.
- Strengthen contingency planning during peak periods to handle high traffic efficiently.



Delay % by Category



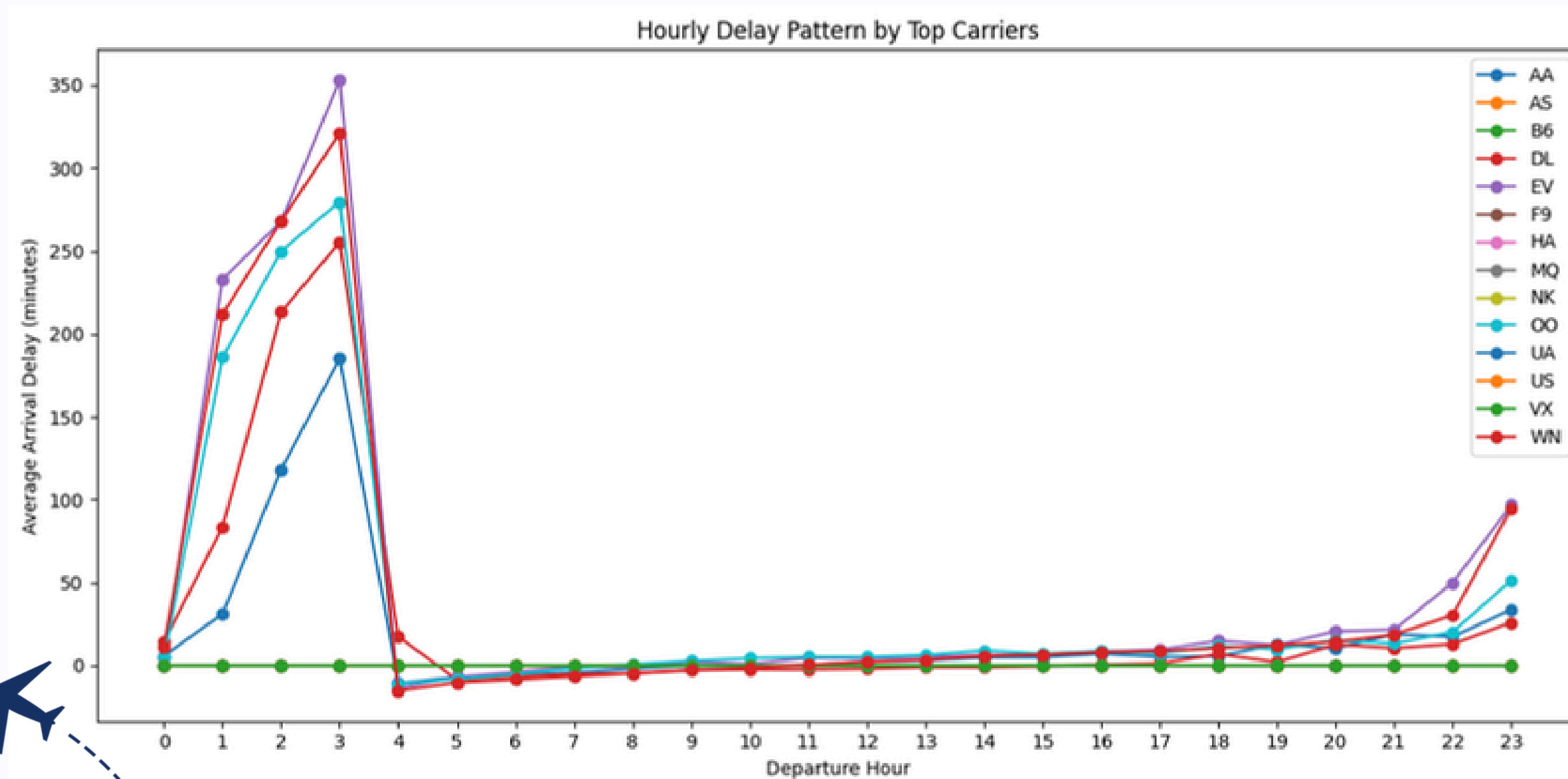
Findings

- Departure and arrival delays together contribute to about 70% of total delays.
- Airline, air system, and late aircraft delays each contribute a moderate share.
- Weather and security delays have minimal impact overall.

Suggestions

- Focus on improving on-time departures through better ground operations and scheduling.
- Strengthen turnaround coordination to reduce arrival-to-departure delay spillover.
- Use data-driven planning to proactively manage airline and air system delays.

Visualizing delays by time of day and Airline



Findings

- Delays peak in early morning (1–4 AM) and rise again late at night.
- Midday hours show relatively stable and lower delays across carriers.

Suggestions

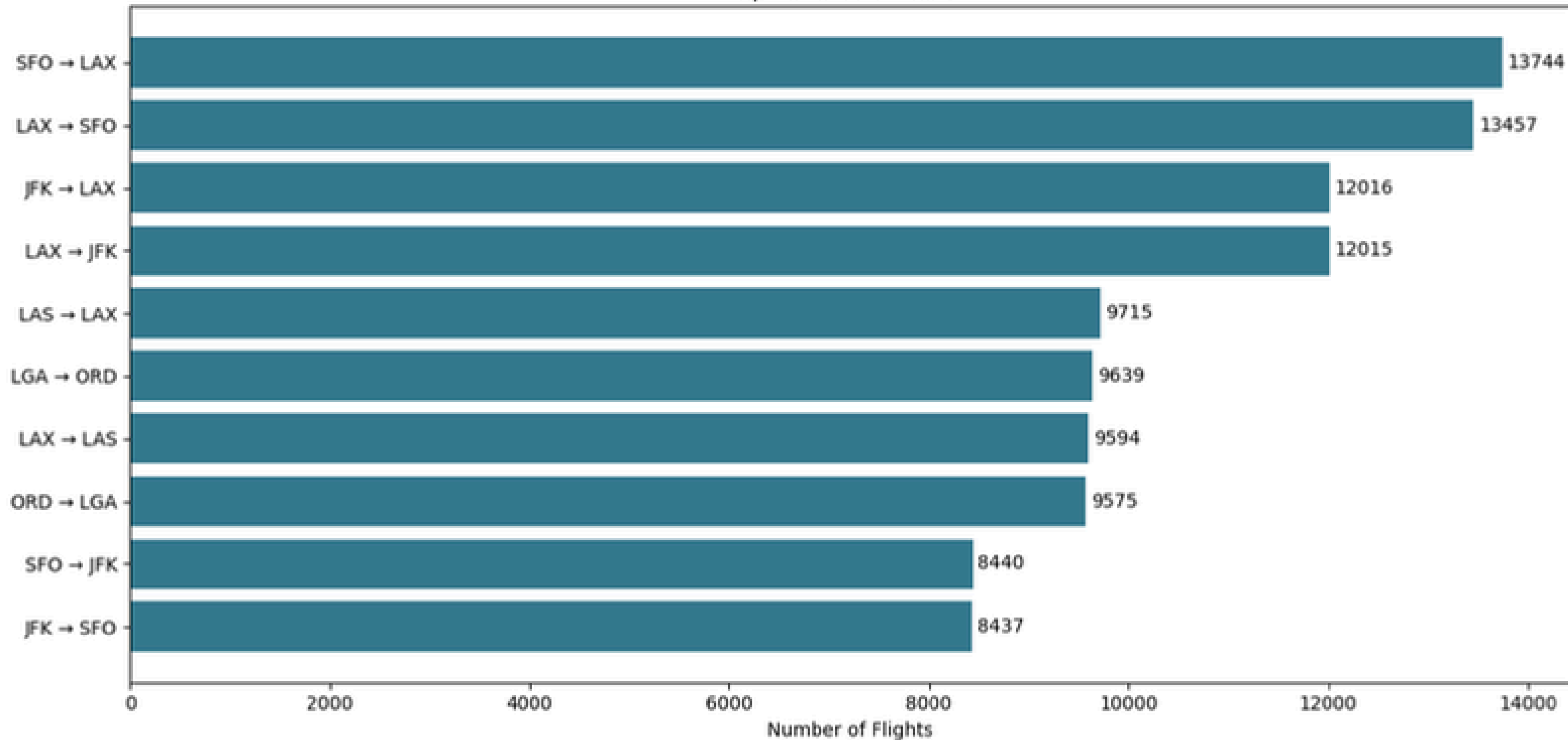
- Strengthen early-morning operations to prevent delay buildup.
- Schedule buffer time and resources for late-night flights to reduce spillover delays.



Top Busiest Routes



Top 10 Busiest Routes



Insights and Suggestions:

- High-traffic routes like LAX–SFO and JFK–LAX dominate operations, indicating strong demand on major hub-to-hub corridors and key revenue-generating routes.
- Optimizing scheduling, capacity, and delay management on these routes can significantly improve efficiency, punctuality, and overall profitability.

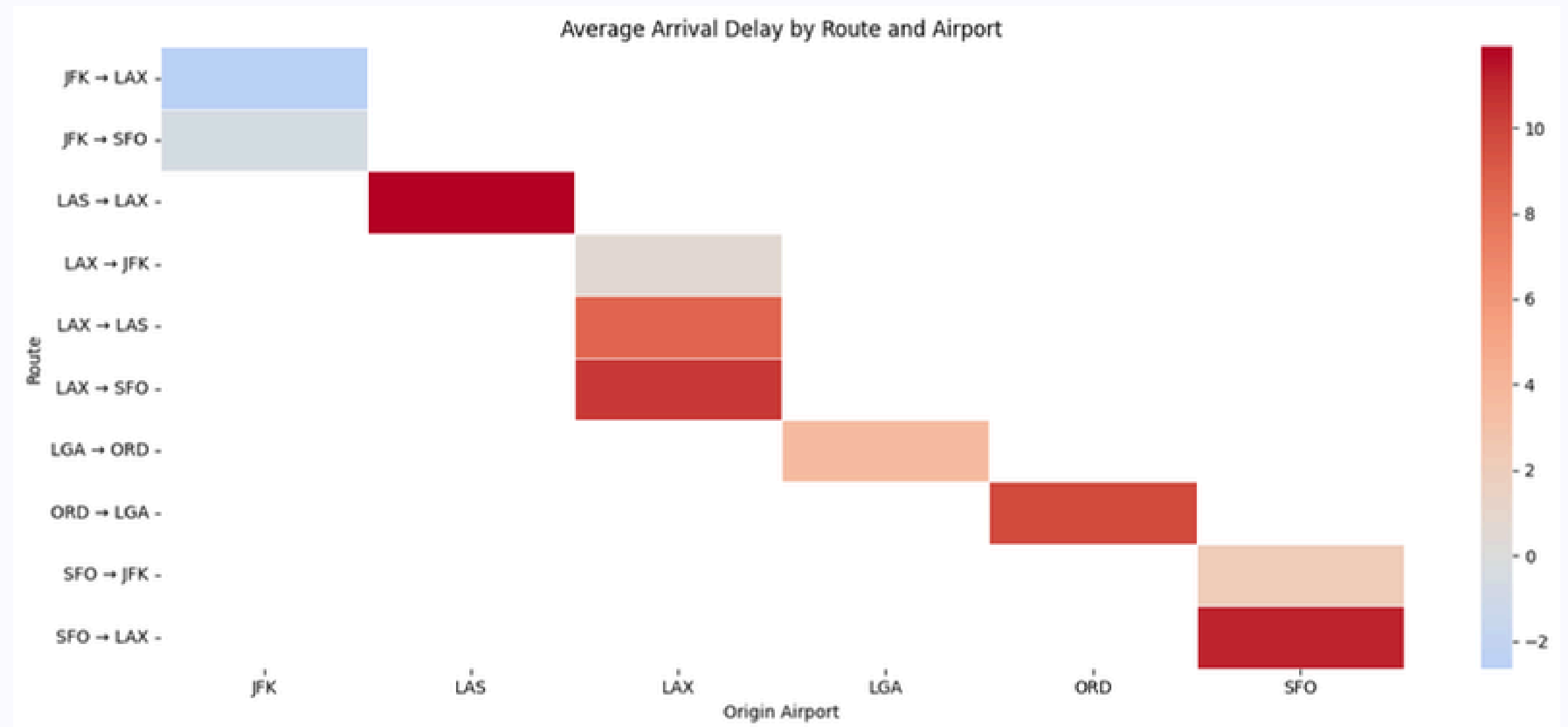
Delay heatmap by Top Routes and Airport

Findings

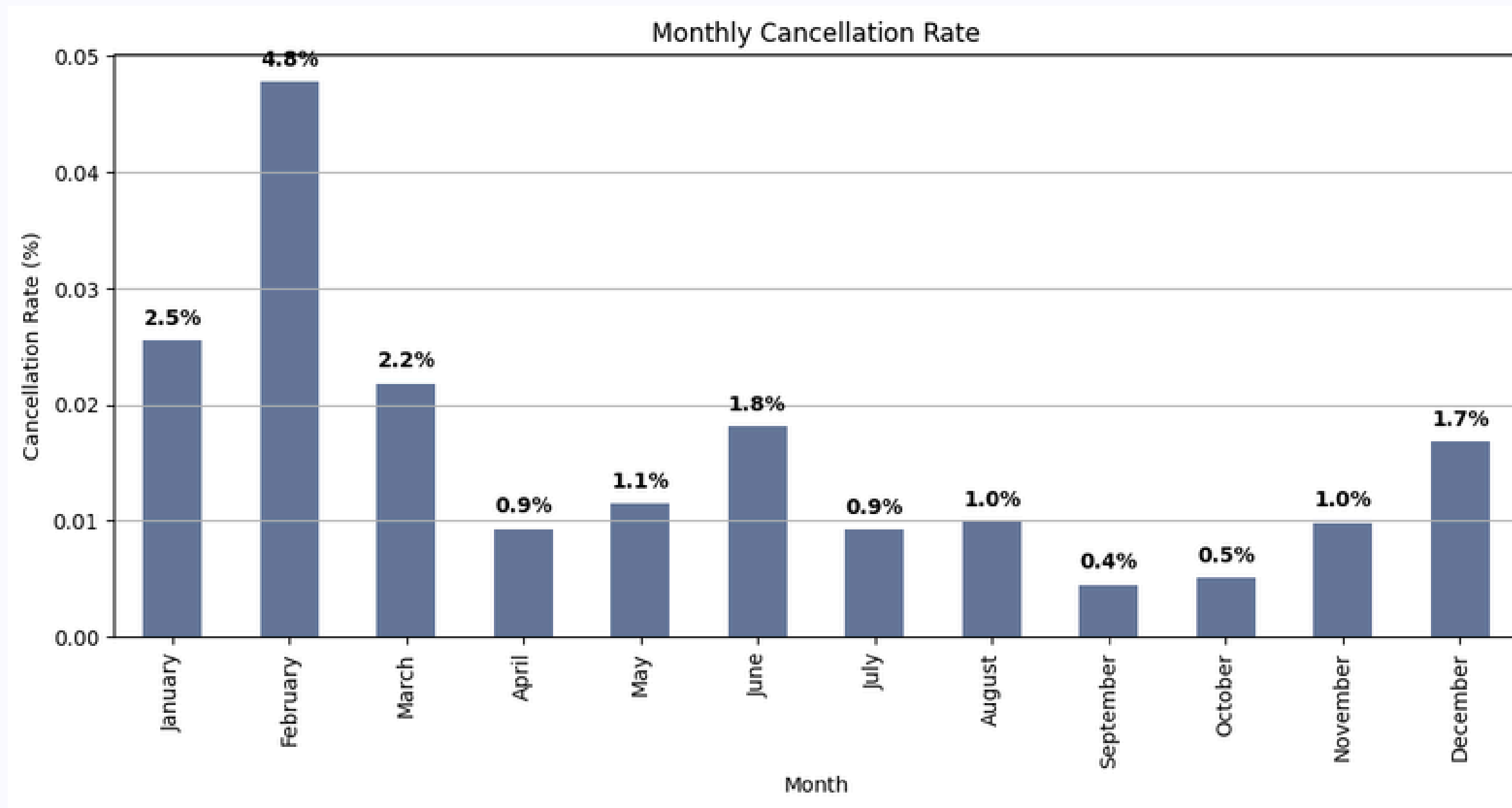
- Routes involving LAX and SFO show higher average arrival delays compared to other routes.
- Some routes like JFK → LAX have lower or near-zero delays, indicating more stable operations.

Suggestions

- Focus operational improvements on high-delay routes connected to LAX and SFO.
- Use successful low-delay routes as benchmarks to improve scheduling and turnaround efficiency.



Monthly Cancellation Trends & Rate



Findings

- Cancellation rates are highest in February and January.
- Rates drop significantly during mid-year, with the lowest in September and October.

Suggestions

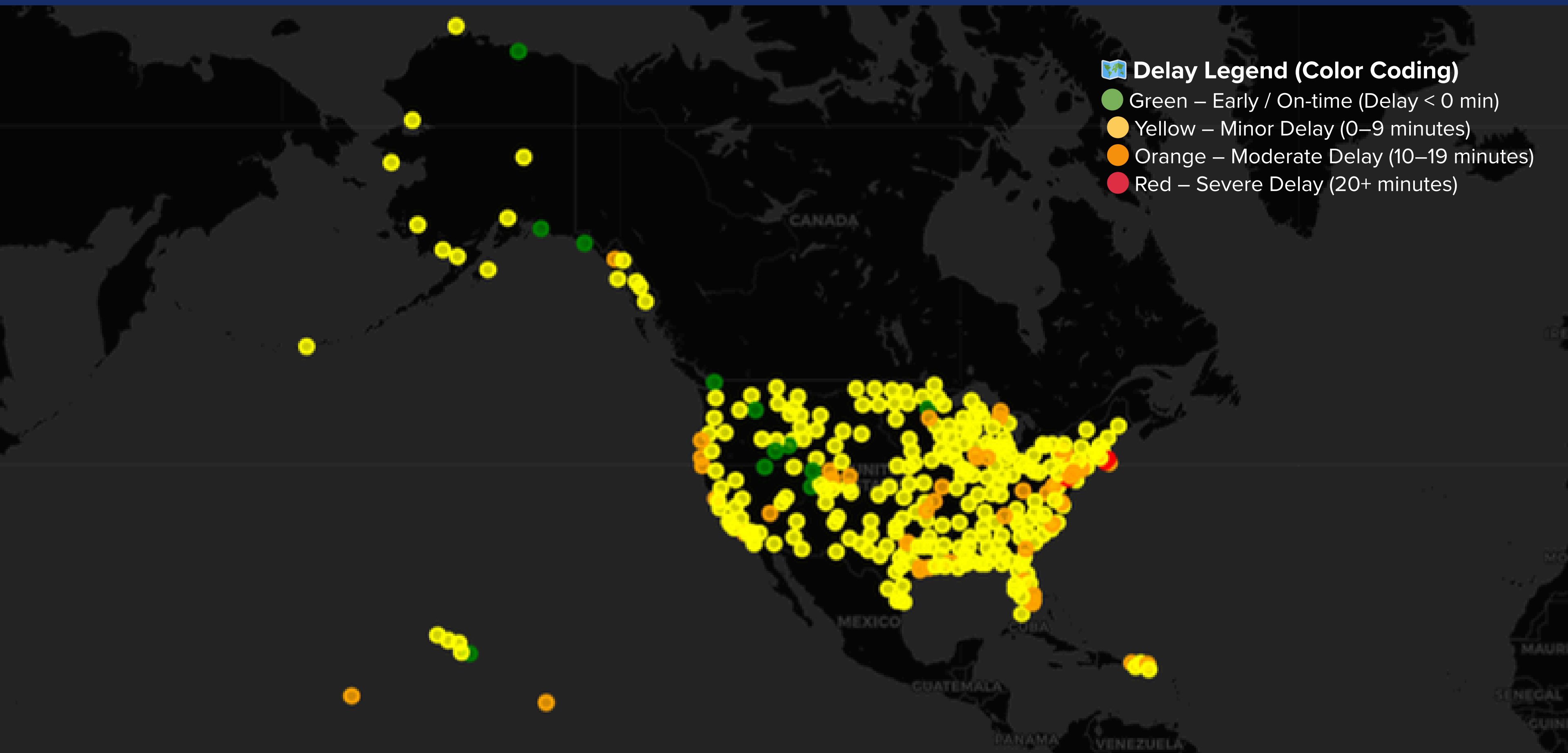
- Strengthen winter weather contingency planning to reduce early-year cancellations.
- Maintain current operations during low-cancellation months and apply best practices year-round.

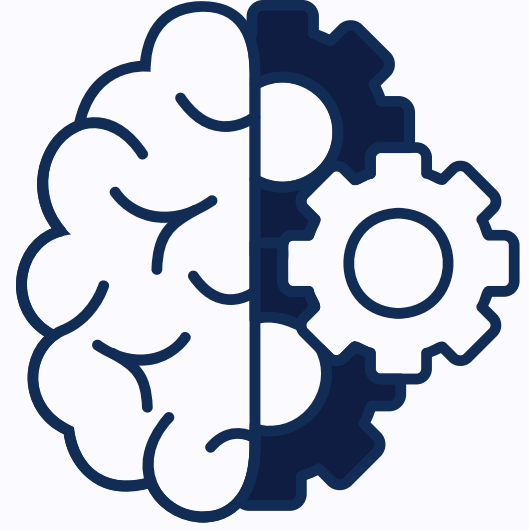




Map Showing Average Delay

-  **Delay Legend (Color Coding)**
-  Green – Early / On-time (Delay < 0 min)
 -  Yellow – Minor Delay (0–9 minutes)
 -  Orange – Moderate Delay (10–19 minutes)
 -  Red – Severe Delay (20+ minutes)





Cancellation Reasons & Percentage



Key Findings

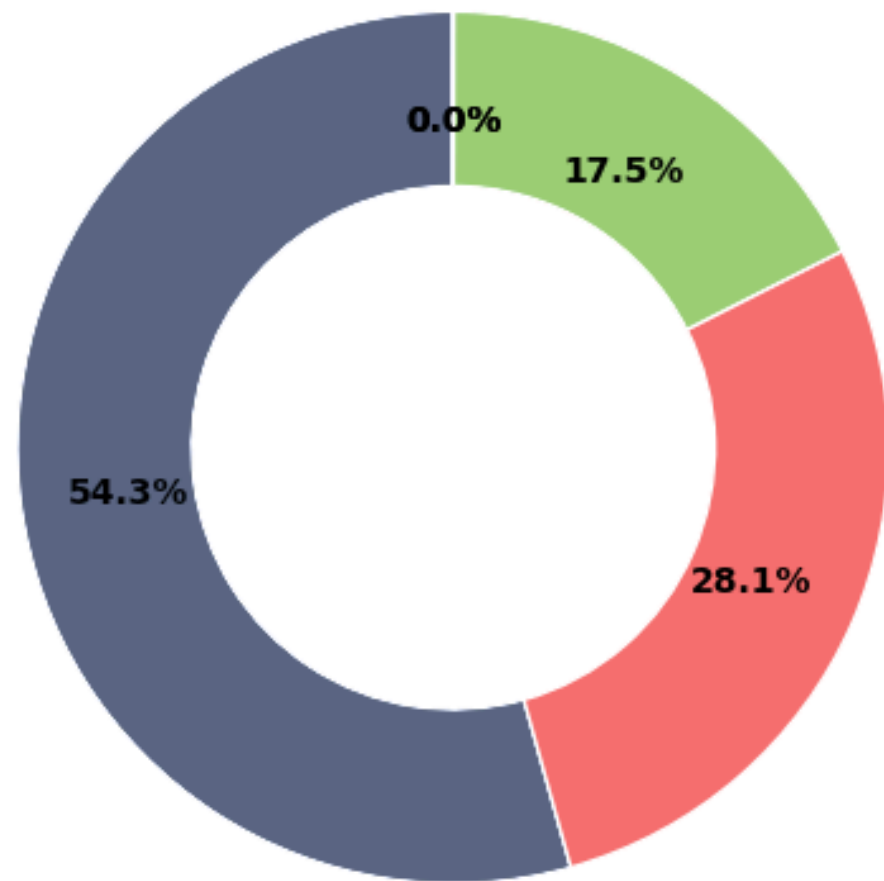
- Weather causes the majority of cancellations (54.3%), making it the biggest risk factor.
- Carrier-related issues (28.1%) are the largest controllable cause.
- NAS delays (17.5%) indicate congestion and airspace dependency.



Business Improvement Suggestions

- Use predictive weather analytics to reschedule flights early and reduce disruptions.
- Improve crew & maintenance planning to minimize carrier-related cancellations.
- Optimize flight scheduling and routing to avoid congested airspaces.

Cancellation Reasons — Donut Chart



Reasons		
Weather	48851	(54.3%)
Carrier	25262	(28.1%)
NAS	15749	(17.5%)
Security	22	(0.0%)
Not Cancelled	0	(0.0%)



Winter Months & Holiday Analysis



Findings

- Winter has the highest cancellation rate (2.9%)
- Spring and Summer show lower and stable cancellations



Suggestions

- Improve winter weather planning
- Add extra buffer flights and crew in winter



Findings

- New Year rush has the highest cancellations (2.55%)
- Christmas and Thanksgiving show moderate impact

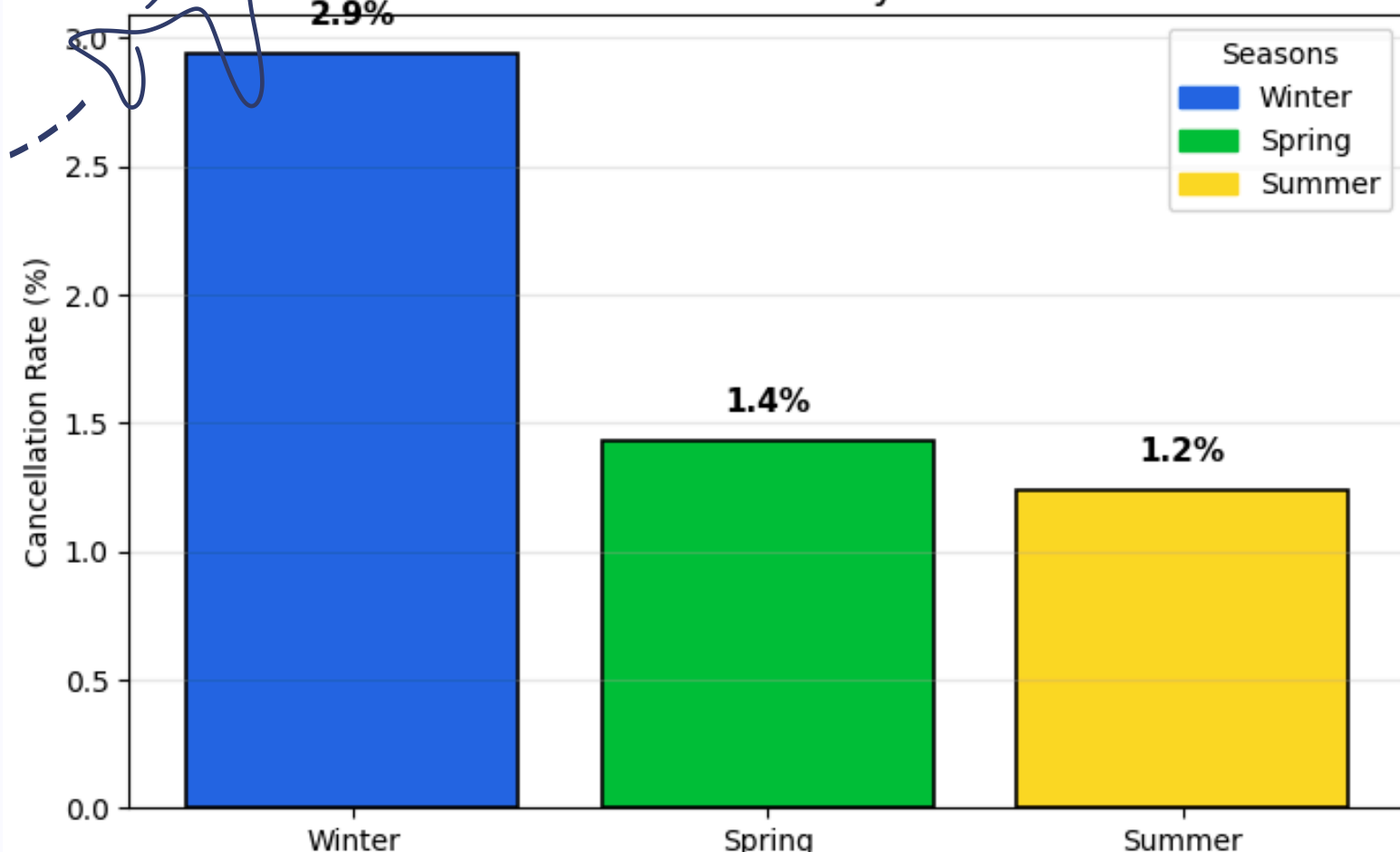


Suggestions

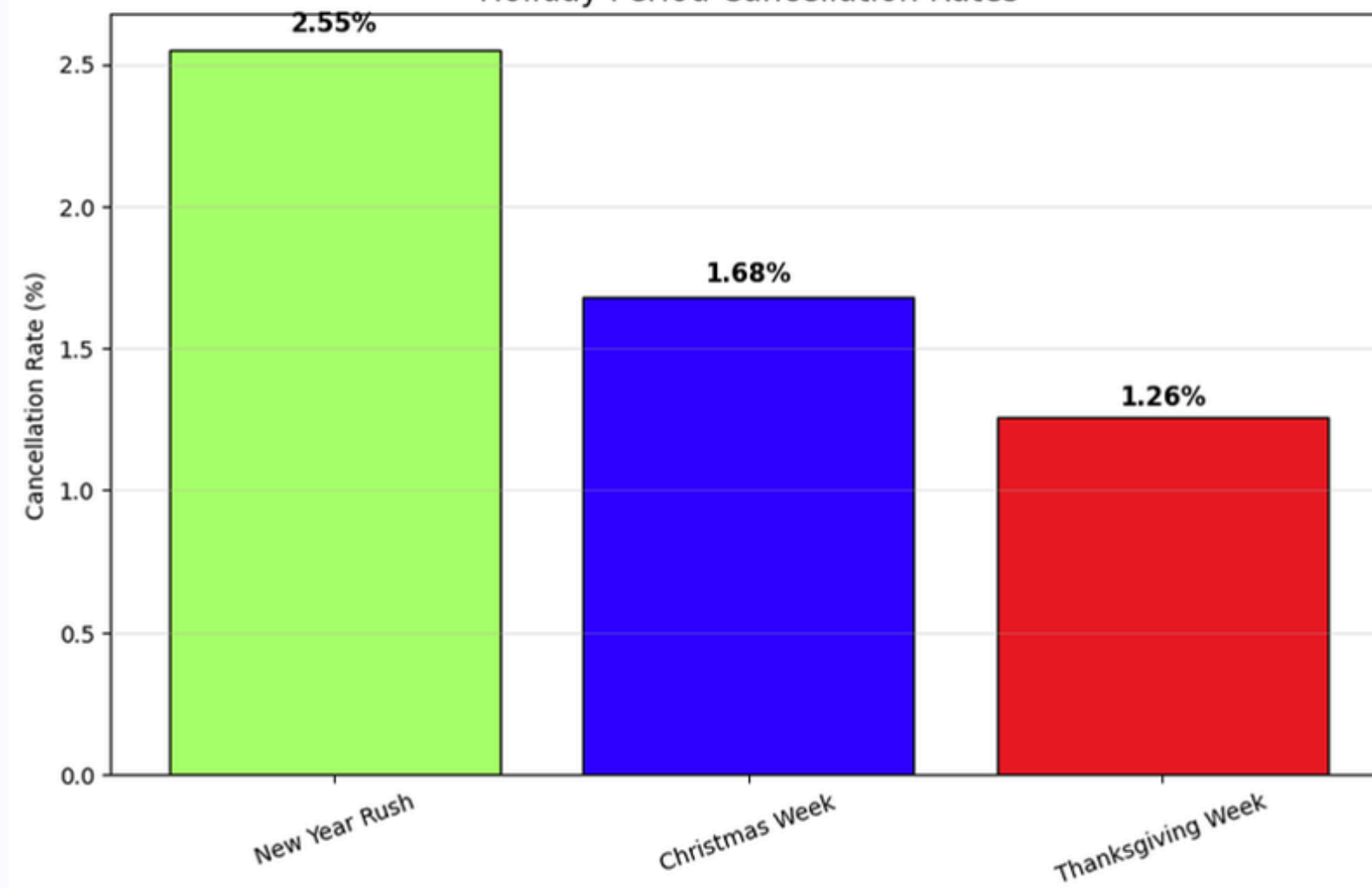
- Increase staffing and backup flights during holidays
- Use advance demand forecasting to reduce disruptions



Cancellation Rate by Season



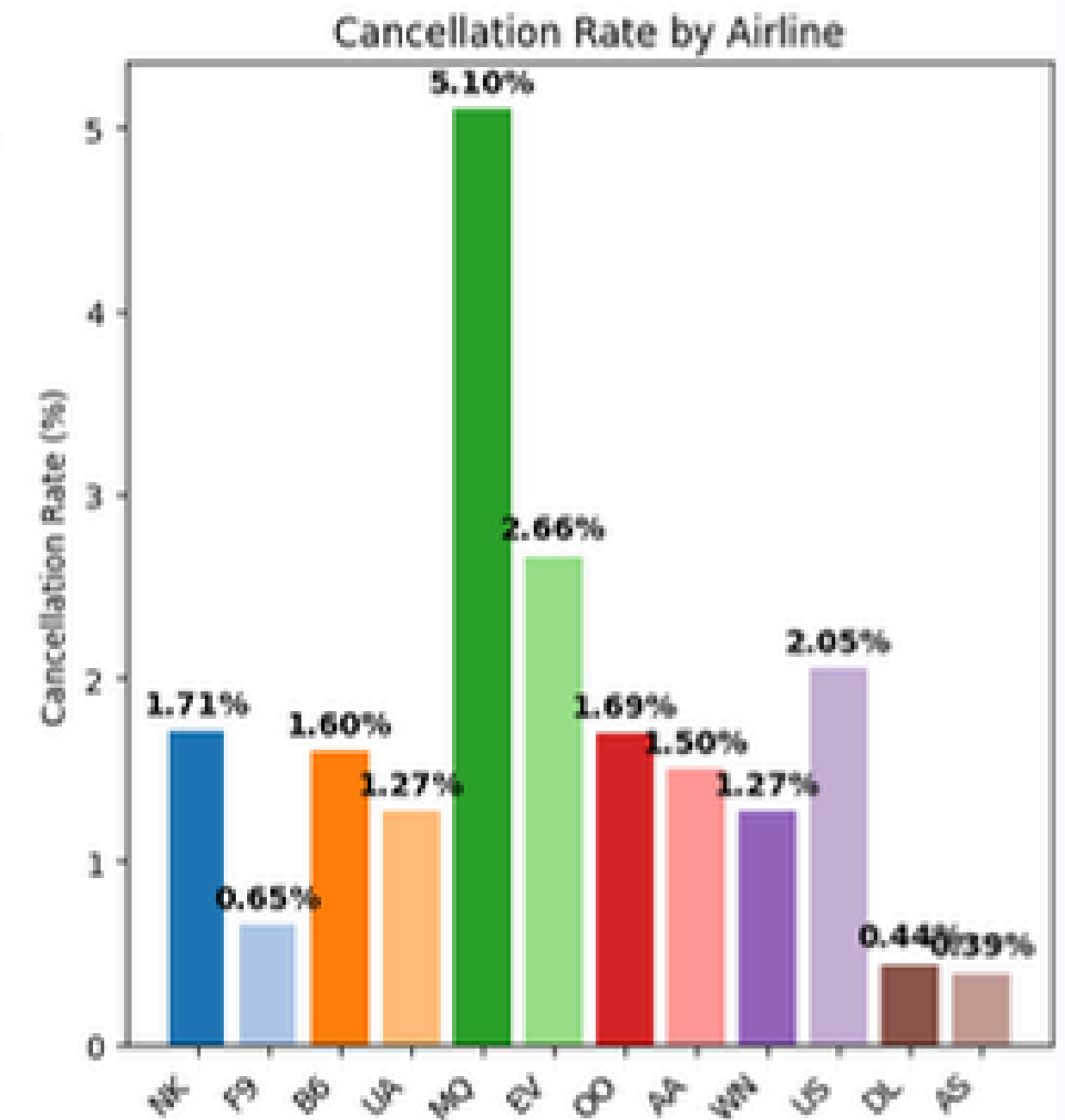
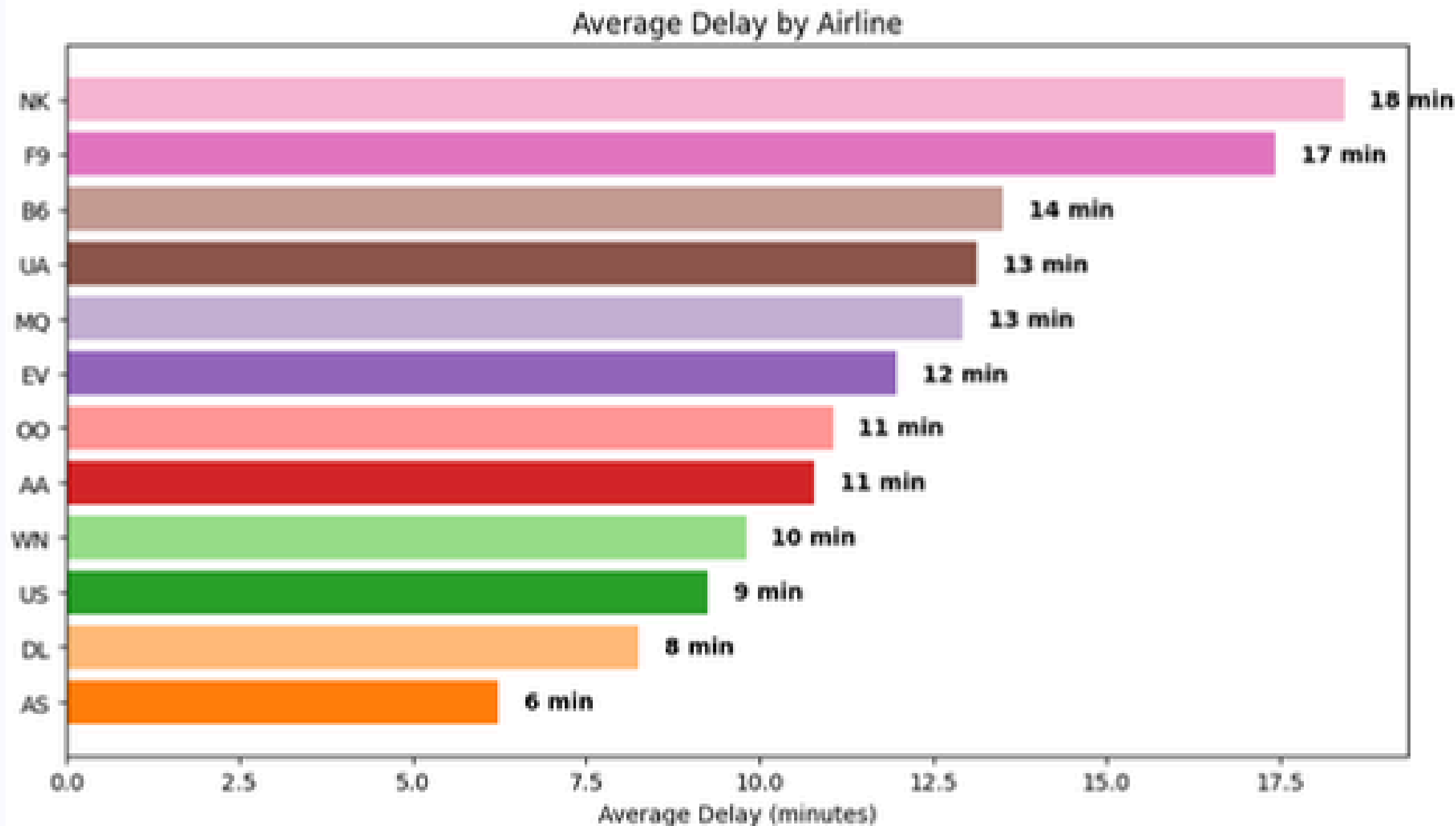
Holiday Period Cancellation Rates





Average Delay & Cancellation Rate by Airline

Airline Performance: Delay & Cancellation



Weather Trend Delay My Month

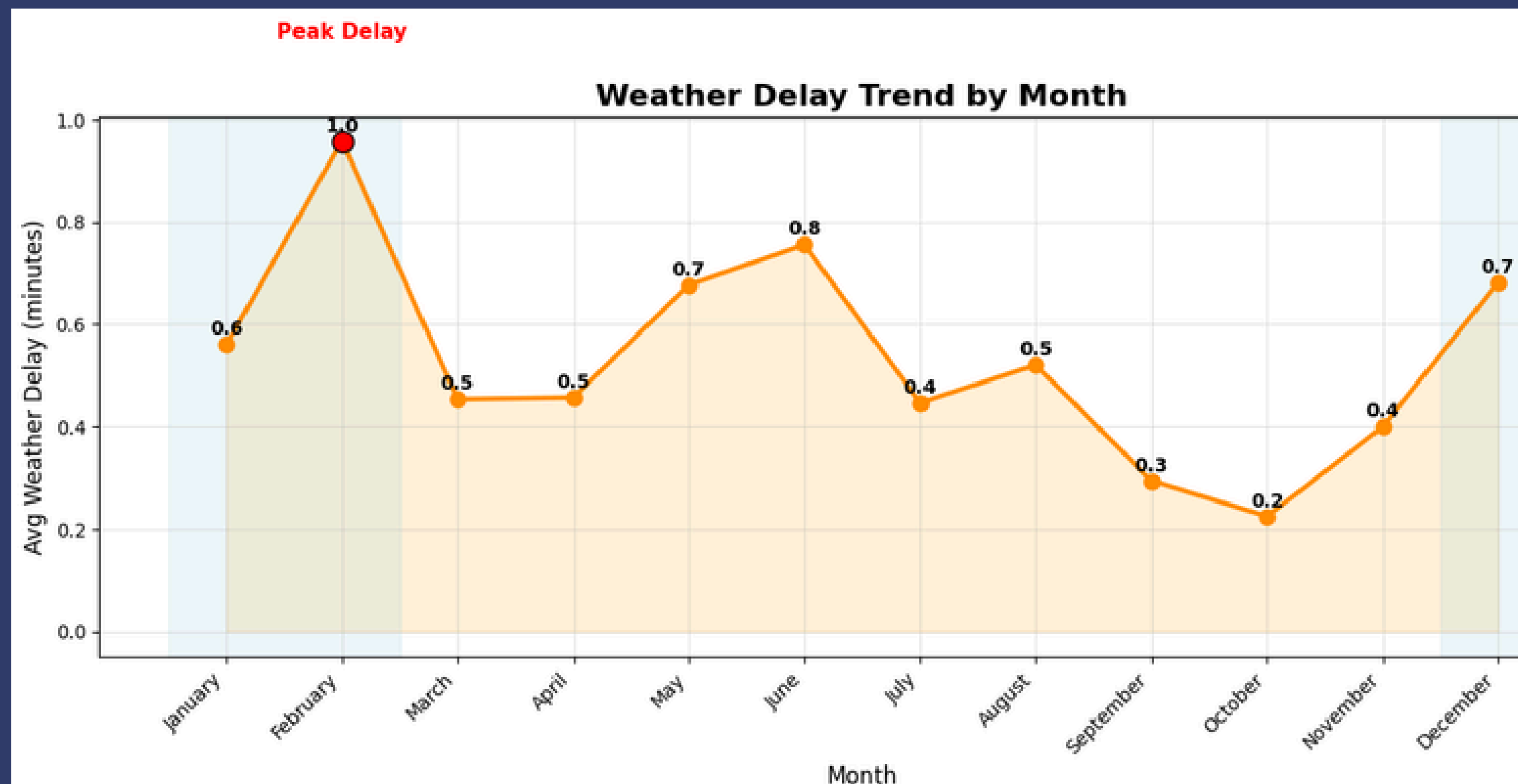


Findings

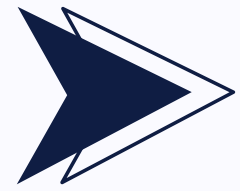
- February shows peak weather delay (1.0 min)
- Lowest delays occur in October (0.2 min)
- Delays rise again during winter months

Suggestions

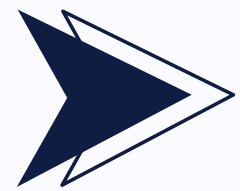
- Strengthen winter weather planning & monitoring
- Add buffer time and contingency flights in peak months



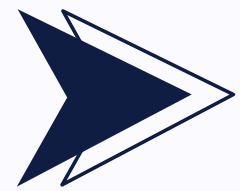
Tech Stack



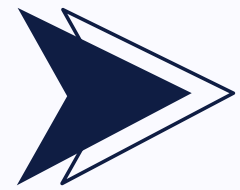
Data Handling: pandas, numpy



Visualization: matplotlib, seaborn, plotly, folium



Optional Dashboard: Streamlit, PowerB, Markdown



Documentation: Jupyter Notebook, PDF, GitHub

Recommendations



Actionable steps to enhance data-driven decision-making and future growth

01 Implement predictive weather monitoring systems

02 Improve crew and aircraft scheduling efficiency

03 Strengthen planning during peak and holiday seasons

04 Use data-driven insights for proactive decision-making

Conclusion

This analysis highlights how data-driven insights can help airlines reduce delays, improve operational efficiency, and enhance customer satisfaction. By leveraging predictive analytics and proactive planning, airlines can achieve more reliable and profitable operations.

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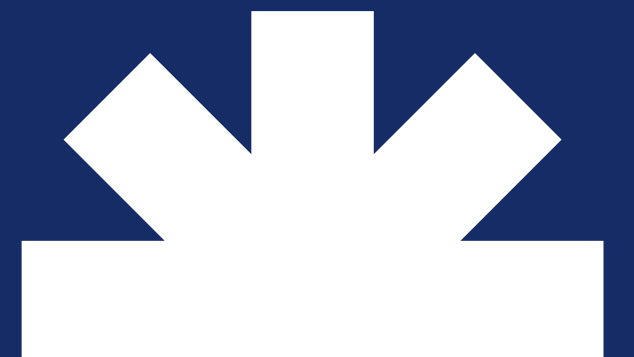
Website

infosysspringboard.com





Any Questions?





Thank
you